

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Design Idea #1

Date: 2/19/2026

Description: The city could use small electric plows that can fit on sidewalks, The plows could clear the snow while it is still snowing, so that it doesn't turn into ice

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____

Date: _____

Class: _____

Design Idea #2

Date: 2/19/2026

Description: The city could install heated sidewalks in important areas and the heat would melt the snow

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____

Date: _____

Class: _____

Design Idea #3

Date: 2/19/2026

Description: The city could create snow teams that would take care of certain neighborhoods. This is a good alternative to property owners clearing their own sidewalks.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____

Date: _____

Class: _____

Design Idea Comparison and Selection (D1, D2)

	weight	plows	Heated sidewalks	teams	truck
Effectiveness	5	5	5	2	3
Safety	5	3	5	3	4
Cost	4	4	2	2	3
Feasibility	3	2	4	2	5
Maintenance and upkeep	3	2	4	2	3
score		68	82	45	71

Design Blueprint (D4)

Design Blueprint #1

Date: 4/13/2026

Sketch:

Manufacturing Methods:

Step 1: Measure and cut wood base

Step 2: Lay and attach wires

Step 3: Place aluminum sheet as conductor

Supply List and Costs:

Item	Cost each	# needed	Total
Wood	Free (scraps from school)	2	\$0
Wires	\$21.49 for yellow \$19.79 for white	2	\$41.28

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Aluminum	14.99	1	\$14.99
Total Cost			\$56.27

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

Week 1:

- Assemble sidewalk prototype
- Finalize design through trial and error

Week 2:

- Test prototype with ice from cafe
- Record results and identify weak points

Week 3:

- Make improvements based on test
- Document final design

